14 Environmental Monitoring and Audit Requirements

14.1 Introduction

14.1.1.1 The findings and recommendations of the EIA report constitute a formal commitment by the Project Proponent to achieve the levels of environmental protection. It also states the Project Proponent's environmental performance criteria. In order to ensure the performance commitments are incorporated throughout various implementation phases (e.g. detailed design, tendering, construction and operation of the Project), a number of contractual, managerial and administrative mechanisms have been implemented, including:

- Setting up of a project organization and hierarchy,
- Development of Environmental Monitoring and Audit (EM&A) programme,
- Outline of Environmental Mitigation Implementation Schedule (EMIS),
- Formulation of Environmental Management Plan (EMP) which includes Waste Management Plan (WMP), and
- Approval of Contractor's Work Method Statement.

14.2 Project Organisation

14.2.1.1 A project organisation consisting of the Independent Environmental Checker (IEC), Contractor's Environmental Team (ET), Engineer's Representative (ER), and Contractor shall be formed to take the responsibilities of the environmental protection for this project.

14.2.1.2 The IEC will be appointed by the Project Proponent to conduct independent auditing on the overall EM&A programme including the implementation of all environmental mitigation, submissions relating to EM&A, and any other submission required under the Environmental Permit (EP). The organisation, responsibilities of respective parties and lines of communication with respect to environmental protection works have been given in the EM&A Manual of the EIA Report.

14.3 EM&A Manual & Implementation Schedule

14.3.1.1 The EM&A is an important aspect in the EIA process that specifies the time frame and responsibilities for the implementation of the environmental mitigation measures identified. Requirements on environmental monitoring (including baseline and impact monitoring) have been given.

14.3.1.2 A project specific EM&A Manual has been prepared based on the latest design information available and EPD's generic EM&A Manual. The Project specific EM&A Manual would specify the following:

- Organisation, hierarchy and responsibilities of the Contractor, the Engineer or ER, ET, and IEC with respect to the EM&A requirements during construction;

- Information on project organisation and programming of construction activities for the Project;
- Requirements with respect to the construction schedule and the necessary EM&A programme to track the varying environmental impact;
- Full details of the methodologies to be adopted, including all field, laboratory and analytical procedures, and details on quality assurance;
- Procedure for undertaking on-site environmental audits;
- Definition of Action and Limit levels;
- Establishment of event and action plans;
- Requirements of reviewing pollution sources and working procedures required in the event of non-compliance of the environmental criteria and complaints;
- Requirements for review of EIA predictions, implementation of mitigation measures, and the effectiveness of the environmental protect and pollution control measures adopted; and
- Presentation requirements for EM&A data and appropriate reporting procedures.

14.3.1.3 An EMIS has also been prepared to summarise all the required mitigation measures that need to be implemented during the design, construction and operation of the proposed project. This EMIS has been included in the EM&A Manual.

14.4 Amendments to EM&A Manual

14.4.1.1 The Contractor shall be requested to review the mitigation measures and EMIS with respect to the design developments and construction methodology. In case when the Contractor needs to update the mitigation measures and the EMIS, an updated EM&A Manual shall be submitted to the EPD for approval. The Contractor shall seek EPD's prior approval on these amendments before construction commences.

14.5 EM&A Programme

14.5.1.1 The Contractor will be requested to implement and operate environmental monitoring programme throughout the entire construction period of the Project. This mechanism will include a system to report the monitoring results on the Project Proponent's website within a period of time, to be agreed by EPD, after the relevant environmental monitoring data are collected. In cases where exceedances are found, the Contractor and ET should take immediate actions to implement remediation measures following the procedures specified in the EM&A Manual

14.5.1.2 Detailed requirements of the EM&A programme are described in the EM&A Manual. Measurements and activities that shall be conducted in accordance with the requirements in the EM&A Manual are summarised as follows:

- Baseline monitoring (on noise, air quality water, ecology etc.),
- Impact monitoring (on noise, air quality water, ecology etc.),

- Remedial actions in accordance with the Event and Action Plan within the time frame in cases where specified criteria in the EM&A Manual are exceeded,
- Logging and keeping records of the details of monitoring results, and
- Preparing and submitting monthly EM&A Reports.

14.6 Environmental Management Plan

14.6.1.1 A systematic EMP shall be set up by the Contractor to ensure effective implementation of the mitigation measures, monitoring and remedial requirements presented in the EIA, EM&A and EMIS. The ER and IEC will audit the implementation status against the EMP and advise the necessary remedial actions required. These remedial actions shall be enforced by the ER through contractual means.

14.6.1.2 The EMP will require the Contractor (together with its sub-contractors) to define in detail how to implement the recommended mitigation measures in order to achieve the environmental performance defined in Hong Kong's environmental legislation and the EIA documentation. The EMP would also need to include a WMP to demonstrate the Contractor's proposal to minimise the waste generation and maximize the re-use of spoil as far as practicable.

14.6.1.3 The review of on-site environmental performance shall be undertaken by the ER and the IEC through a systematic checklist and audit once the construction commences. The environmental performance review programme comprises a regular assessment on the effectiveness of the EMP. Reference should be made to ETWBTC 19/2005 "Environmental Management on Construction Sites" or its latest versions, and any other relevant Technical Circulars.

14.7 Method Statements

14.7.1.1 The environmental aspects of working methods will be controlled through the checking of the Contractor's method statements which will be submitted and approved by the ER prior to the works being carried out. The Project Proponent will specify an arrangement whereby method statements will be scrutinized and signed off by ER before being approved. This will ensure that the environment is consistently and routinely considered in all works processes.